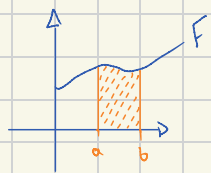


Lecture 16: Double Integrals

Remember: Function of 1 variable

$$\int_a^b f(x) dx = \text{area below graph of } f \text{ over } [a, b].$$



Double integral = volume below graph $z = f(x, y)$ over a region R in xy -plane

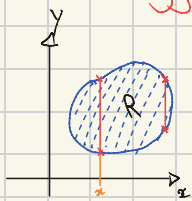
$$\iint_R f(x, y) dA \quad \text{"Area"}$$

Definition: cut R into small pieces of Area ΔA_i ; $(\sum_{i=1}^n f(x_i, y_i) \cdot \Delta A_i)$ Take the limit as $\Delta A_i \rightarrow 0$, get $\iint$

To compute $\iint_R f(x, y) dA$: take slices

Let $S(x)$ = area of slice by plane $\parallel yz$ -plane. Then volume = $\int_{x_{\min}}^{x_{\max}} S(x) dx$

For given x , $S(x) = \int_{y_{\min}(x)}^{y_{\max}(x)} f(x, y) dy$



$$\iint_R f(x, y) dA = \int_{x_{\min}}^{x_{\max}} \left[\int_{y_{\min}(x)}^{y_{\max}(x)} f(x, y) dy \right] dx$$

Iterated Integral

$x_{\min}$ and $x_{\max}$ are NUMBERS; $y_{\min}(x)$ and $y_{\max}(x)$ are functions of x

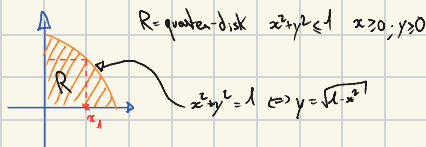
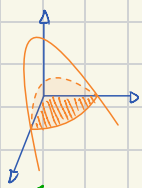
Example 1: $z = 1 - x^2 - y^2$ region: $0 \leq x \leq 1$
 $0 \leq y \leq 1$

$$\int_0^1 \int_0^1 1 - x^2 - y^2 dy dx$$

① Inner Integral: $\int_0^1 (1 - x^2 - y^2) dy = \left[y - x^2 y - \frac{y^3}{3} \right]_0^1 = (1 - x^2 - \frac{1}{3}) - 0 = \frac{2}{3} - x^2$

② Outer Integral: $\int_0^1 (\frac{2}{3} - x^2) dx = \left[\frac{2}{3}x - \frac{1}{3}x^3 \right]_0^1 = \frac{1}{3}$

Example 2: Same function



For given x , the range of y is from $y=0$ to $\sqrt{1-x^2}$

$$\iint_R 1 - x^2 - y^2 dA = \int_0^1 \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2) dy dx$$

Inner: $\int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2) dy = \left[y - x^2 y - \frac{y^3}{3} \right]_0^{\sqrt{1-x^2}} = \sqrt{1-x^2} - x^2 \sqrt{1-x^2} - \frac{(1-x^2)^{3/2}}{3} = (1-x^2)\sqrt{1-x^2} - \frac{(1-x^2)^{3/2}}{3} = (1-x^2)^{1/2} \left[1 - \frac{(1-x^2)}{3} \right] = (1-x^2)^{3/2} \left[\frac{3 - (1-x^2)}{3} \right] = \frac{2}{3} (1-x^2)^{3/2}$

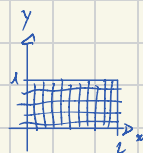
Outer: $\int_0^1 \frac{2}{3} (1-x^2)^{3/2} dx = \int_0^{\pi/2} \frac{2}{3} \cos^2 \theta \cdot \cos \theta d\theta = \frac{2}{3} \int_0^{\pi/2} \cos^3 \theta d\theta$
 $x = \sin \theta; (1-x^2)^{1/2} = \cos \theta$
using $\sin^2 \theta + \cos^2 \theta = 1$

$$= \frac{2}{3} \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right) \cos \theta d\theta = \frac{2}{3} \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2} \cos 2\theta + \frac{1}{2} \cos^2 2\theta \right) d\theta = \frac{\pi}{8}$$

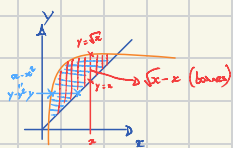
 $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

Will be easier in polar coordinates
Exchanging order of integration

Example 1: $\int_0^1 \int_0^2 y dy dx = \int_0^2 \int_0^1 y dx dy$



Example 2: $\int_0^1 \int_x^{\sqrt{x}} \frac{e^y}{y} dy dx$
 $= \int_0^1 \int_{y^2}^y \frac{e^y}{y} dx dy$



Inner: $\left[\frac{e^y}{y} x \right]_{x=y^2}^x = e - e^y$ Outer: $\int_0^1 e^y - ye^y dy = [ye^y - e^y]_0^1 = e - 2$